NEW PROPOSED RESEARCH QUESTION:

Radiomics-Based Prediction of Clinical Outcomes in Pediatric High-Grade Gliomas Using TCIA MRI Data

DATASET:
The Cancer Imaging Archive
DFCI-BCH-BWH-PEDs-HGG | MR imaging of pediatric subjects with high-grade gliomas
https://www.cancerimagingarchive.net/collection/DFCI-BCH-BWH-PEDs-HGG/

Task	Lead(s)	Collaborators and Reviewers	What does the whole team need to understand
Literature review and Background Research	Sherri	Atheena	The clinical importance of pediatric high-grade gliomas, current imaging-based prognostic methods, and how radiomics can improve outcome prediction.
Dataset interpretation	Atheena	Sherri	The structure and metadata of the TCIA DFCI-BCH-BWH-PEDs-HGG dataset, available MRI sequences, and outcome variables.
Dataset cleaning/processing	Shruti	Ria	Preprocessing workflow: data conversion, skull stripping, bias correction, registration, normalization, and segmentation consistency across cases.
Analysis (statistics, machine learning, etc.)	Ria	Shruti	The radiomics feature extraction process, feature selection

			(LASSO/PCA), and ML model training for survival or outcome prediction.
Validation and Evaluation	Shruti	Ria	How model performance will be assessed (AUC, accuracy, sensitivity, specificity, F1 score) and how overfitting will be prevented.
Interpretation	Sherri	Atheena	How radiomic features relate to clinical outcomes, biological plausibility, and how to discuss limitations and uncertainty.
Documentation and Reproducibility	Atheena	Sherri	How the GitHub repository is organized, how preprocessing and analysis steps can be reproduced, and how transparency is maintained in reporting.

Children diagnosed with pediatric high-grade gliomas (HGG) experience poor survival outcomes and limited treatment options, which matters because these aggressive brain tumors account for a disproportionate share of childhood cancer mortality and lack reliable imaging-based prognostic tools. Previous research has shown that radiomic features from MRI can predict survival and molecular subtypes in adult gliomas, but no study has systematically evaluated whether MRI-derived radiomic signatures can predict clinical outcomes in pediatric HGG using open, multi-sequence data from TCIA.